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# Some trigonometrical inequalities with applications to the theory of series.

By

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1. This paper is primarily concerned with Dirichlet's series of the type

$$f(s) = f(\sigma + ti) = \sum_1^\infty a_n e^{-\lambda_n s} = \sum_1^\infty a_n l_n^{-s} \qquad (l_n = e^{\lambda_n}) \tag{1}$$

n which the exponents $\lambda_n$ satisfy

$$\lambda_n - \lambda_{n-1} \geqq \gamma > 0 \qquad (n = 2, 3, \ldots).$$

The first aim is to obtain inequalities to take the place of the familiar relationships which hold between the coefficients $a_n$ and the function $f(s)$ in the special case $\lambda_n = n\gamma$, when the series is a power series in $e^{-\gamma s}$ and $f(s)$ a periodic function with period $2\pi i/\gamma$. The basic inequalities may be stated, without essential loss of content, for trigonometrical polynomials (with exponents $\lambda_n$). These inequalities (Theorems 1 and 2) are set out in § 2 and § 4.

The paper arises out of an attempt to obtain a satisfactory analogue, in the sense explained in a recent paper[1]), of the well known Tauberian theorem of Hardy and Littlewood for 'high indices'. Theorem 1 is applied, in § 5, to the proof of such an analogue (Theorem 3), and, in § 6, to the proof of a theorem of Pólya on the existence of singular points on the line of convergence of Dirichlet's series of the above type. A simple application to a special class of almost periodic functions follows in § 7.

Some of the results of this paper have affinities with recent work of N. Wiener and Paley, but the point of view is different and it is not

[1]) Ingham, 1 (particularly 464—465 and 479,2)).

I take this opportunity of correcting and amplifying the allusions to the work of Karamata in the footnote to p. 465 of this paper, which are in certain respects inaccurate or misleading. In the first place, the words 'the case $\theta = 1$ of' in the third line from the bottom of the page were inserted through inadvertence and should be deleted. [In deducing Karamata's Satz B′ (with specialised conditions on $G(s)$) from Theorem I of my paper we take $\theta = T$]. Secondly there should have been a reference to Karamata's note 1, which contains a very clear statement of the analogy between the two classes of Tauberian theorems, at any rate for theorems of the type represented by Theorem III of my paper.

easy to analyse the connections in detail[2]). Here no systematic use is made of the Lebesgue integral, and the results are in this respect much less general than those of Wiener and Paley. On the other hand, theorems are here given in which the lower bound ($\pi$) of certain relevant classes of numbers is determined exactly (though it remains undecided, except in one case, whether this bound is attained or not). The question of the 'best possible' nature of the results is discussed in § 3 and § 8.

2. **Theorem 1.** *Let*

$$f(t) = \sum_{n=N}^{N'} a_n e^{-\lambda_n t i} = \sum_n a_n e^{-\lambda_n t i}, \tag{2}$$

*where the $\lambda_n$ are real and satisfy*

$$\lambda_n - \lambda_{n-1} \geqq \gamma > 0 \qquad (N < n \leqq N'), \tag{3}$$

*and let*

$$T = \frac{\pi + \varepsilon}{\gamma} > \frac{\pi}{\gamma}.$$

*Then*

$$|a_n| \leqq \frac{\mathfrak{A}(\varepsilon)}{2T} \int_{-T}^{T} |f(t)|\, dt \qquad (N \leqq n \leqq N'), \tag{4}$$

$$\sum_n |a_n|^2 \leqq \frac{\mathfrak{A}(\varepsilon)}{2T} \int_{-T}^{T} |f(t)|^2\, dt, \tag{5}$$

*where $\mathfrak{A}(\varepsilon)$ is a function of $\varepsilon$ only, which has a finite value for every $\varepsilon > 0$, and which may be chosen so that $\varepsilon \mathfrak{A}(\varepsilon) \to \mathfrak{A}$ as $\varepsilon \to +0$, where $\mathfrak{A}$ is a finite absolute constant.*

*There are corresponding results for infinite series converging uniformly over the interval $-T \leqq t \leqq T$.*

The last part is an immediate extension, and we may confine ourselves to trigonometrical polynomials of the form (2). We may take $\gamma$ and $T$ to be any convenient positive numbers whose product is $\pi + \varepsilon$, since the substitution $t \to ct$ $(c > 0)$ merely changes $\gamma$ to $\gamma c$ and $T$ to $T/c$.

Let $k(t)$ be any function absolutely integrable over $(-\infty, \infty)$ and let

$$K(u) = \int_{-\infty}^{\infty} k(t) e^{-ut i}\, dt. \tag{6}$$

Then

$$\int_{-\infty}^{\infty} k(t) f(t) e^{\lambda_n t i}\, dt = \sum_m a_m K(\lambda_m - \lambda_n), \tag{7}$$

$$\int_{-\infty}^{\infty} k(t) |f(t)|^2\, dt = \sum_{m,n} a_m \bar{a}_n K(\lambda_m - \lambda_n). \tag{8}$$

[2]) Wiener, 1; Paley and Wiener, 1, Chapter VII.

Take $T = \pi$, $\gamma = 1 + \delta$ $(\delta = \varepsilon/\pi)$, and

$$k(t) = \begin{cases} \cos \frac{1}{2} t & (|t| \leqq \pi), \\ 0 & (|t| > \pi). \end{cases}$$

Then

$$K(u) = \frac{4 \cos u\pi}{1 - 4u^2}.$$

Since, by (3), $|\lambda_m - \lambda_n| \geqq |m-n|\gamma = |m-n|(1+\delta) > 1$ $(m \neq n)$ we have (the dash indicating the omission of the term $m = n$)

$$\sum_m{}' |K(\lambda_m - \lambda_n)| \leqq \sum_m{}' \frac{4}{4(m-n)^2\gamma^2 - 1} < \frac{8}{\gamma^2} \sum_{r=1}^{\infty} \frac{1}{4r^2 - 1} \tag{9}$$

$$= \frac{4}{\gamma^2} \sum_{r=1}^{\infty} \left( \frac{1}{2r-1} - \frac{1}{2r+1} \right) = \frac{4}{\gamma^2} = \frac{K(0)}{(1+\delta)^2}.$$

Taking $n = \nu$ in (7), where $|a_\nu|$ is the greatest $|a_n|$, we deduce

$$\left| \int_{-\infty}^{\infty} k(t) f(t) e^{\lambda_\nu t i} dt \right| \geqq |a_\nu K(0)| - |a_\nu| \sum_m{}' |K(\lambda_m - \lambda_\nu)|,$$

whence, by (9) and the definition of $k(t)$,

$$\int_{-\pi}^{\pi} |f(t)| dt \geqq |a_\nu| \left( 4 - \frac{4}{(1+\delta)^2} \right) = \max |a_n| \cdot \frac{4\delta(2+\delta)}{(1+\delta)^2},$$

and this is (4) with

$$\mathfrak{A}(\varepsilon) = \frac{2\pi(1+\delta)^2}{4\delta(2+\delta)} = \frac{\pi(\pi+\varepsilon)^2}{2\varepsilon(2\pi+\varepsilon)}.$$

Again, since $2|a_m \bar{a}_n| \leqq |a_m|^2 + |a_n|^2$ and $K(\lambda_m - \lambda_n) = K(\lambda_n - \lambda_m)$, we deduce from (8) that

$$\int_{-\infty}^{\infty} k(t) |f(t)|^2 dt = \sum_n |a_n|^2 K(0) + \vartheta \sum_{m,n}{}' \frac{|a_m|^2 + |a_n|^2}{2} |K(\lambda_m - \lambda_n)|$$

$$= \sum_n |a_n|^2 \left( K(0) + \vartheta \sum_m{}' |K(\lambda_m - \lambda_n)| \right) \quad (|\vartheta| \leqq 1), \tag{10}$$

whence, by (9) and the definition of $k(t)$,

$$\int_{-\pi}^{\pi} |f(t)|^2 dt \geqq \sum_n |a_n|^2 \left( 4 - \frac{4}{(1+\delta)^2} \right) = \frac{4\delta(2+\delta)}{(1+\delta)^2} \sum_n |a_n|^2,$$

which gives (5) with the same $\mathfrak{A}(\varepsilon)$ as before.

The function $k(t)$ may be chosen in various ways without affecting the principle of the above argument. Thus we may take $\gamma = \pi + \varepsilon$, $T = 1$, and

$$k(t) = \begin{cases} 1 - t^2 & (|t| \leqq 1) \\ 0 & (|t| > 1), \end{cases}$$

which gives

$$K(u) = \frac{4(\sin u - u\cos u)}{u^3}.$$

A suitable analogue of (9) may be derived from the inequality

$$|uK(u)| \leqq |r\pi K(r\pi)| \qquad (|u| \geqq r\pi;\ r \text{ any positive integer})$$

(which is easily verified), and we deduce (4) and (5) with

$$\mathfrak{A}(\varepsilon) = \frac{3(\pi+\varepsilon)}{2\varepsilon}.$$

There is another argument, based on a different idea, which is of interest, though it seems to apply only to (5). Take $\gamma = 1$, $T = \pi + \varepsilon$, and

$$k(t) = k_\varepsilon(t) = \frac{1+\cos t}{(\pi^2 - t^2)^2}\{(\pi+\varepsilon)^2 - t^2\} = 2\left(\frac{\cos\frac{1}{2}t}{\pi^2 - t^2}\right)^2\{(\pi+\varepsilon)^2 - t^2\}$$

(an integral function of $t$). Then $K(u) = K_\varepsilon(u)$ vanishes for $|u| \geqq 1$, as may be proved by direct (real variable) calculation, or (more simply) by an application of Cauchy's theorem to a semicircular contour with its diameter along the real axis in the $t$-plane. Thus (8) gives

$$(11) \qquad K_\varepsilon(0)\sum_n |a_n|^2 = \int_{-\infty}^{\infty} k_\varepsilon(t)\,|f(t)|^2\,dt \leqq \int_{-\pi-\varepsilon}^{\pi+\varepsilon} k_\varepsilon(t)\,|f(t)|^2\,dt,$$

since $k_\varepsilon(t) \leqq 0$ outside the interval $(-\pi-\varepsilon, \pi+\varepsilon)$. Now in this interval we have

$$k_\varepsilon(t) = \frac{1}{2(\pi+|t|)^2}\left(\frac{\sin\frac{1}{2}(\pi-|t|)}{\frac{1}{2}(\pi-|t|)}\right)^2\{(\pi+\varepsilon)^2 - t^2\} \leqq \frac{(\pi+\varepsilon)^2}{2\pi^2}.$$

Also

$$K_\varepsilon(0) = \frac{\varepsilon(2\pi+\varepsilon)}{2\pi},$$

as is easily verified by (real or complex variable) calculation. Hence (11) implies (5) with

$$\mathfrak{A}(\varepsilon) = \frac{2(\pi+\varepsilon)^3}{\pi\varepsilon(2\pi+\varepsilon)}.$$

3. By considering the (periodic) function

$$(1+e^{-s})^{-k} = \sum_0^\infty a_n e^{-n\sigma}e^{-nti} \quad (s = \sigma + ti,\ \sigma > 0),$$

where $k$ is an integer greater than 1, and making $\sigma \to +0$, we see at once that the $\pi$ of Theorem 1 cannot be replaced by any smaller absolute constant. A more delicate question is whether the $\varepsilon$ can be replaced by 0 and $\mathfrak{A}(\varepsilon)$ by an absolute constant $\mathfrak{A}$ (as in the case of periodic functions). We return to this in § 8, where we explain also the general principles which govern the choice of $k(t)$ in the foregoing proofs.

4. The inequality (5) corresponds to Bessel's inequality for periodic functions. There is an inequality in the opposite sense constituting with (5) a counterpart of the Parseval identity. This, however, is easier, and the number $\pi$ does not play the same critical rôle.

**Theorem 2.** *If $f(t)$ satisfies the conditions of Theorem* 1, *and if*

$$T = \frac{\tau}{\gamma} > 0,$$

*then*

$$\sum_n |a_n|^2 \geqq \frac{\mathfrak{a}(\tau)}{2T} \int_{-T}^{T} |f(t)|^2 \, dt,$$

*where $\mathfrak{a}(\tau)$ is a function of $\tau$ only which is $> 0$ for every $\tau > 0$.*

For convenience we use the formulae of the first proof of Theorem 1, though in fact the function $k(t)$ need not be chosen with the same care.

If $\tau \geqq \pi$, we take $T = \frac{1}{2}\pi$ (as we may), so that $\gamma \geqq 2$. Then from (10) and (9) we deduce

$$\int_{-\pi}^{\pi} \cos \tfrac{1}{2} t \, |f(t)|^2 \, dt \leqq \sum_n |a_n|^2 \left(4 + \frac{4}{\gamma^2}\right) \leqq 5 \sum_n |a_n|^2,$$

whence

$$\sum_n |a_n|^2 \geqq \frac{1}{10} \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} |f(t)|^2 \, dt = \frac{\pi}{10} \cdot \frac{1}{2T} \int_{-T}^{T} |f(t)|^2 \, dt.$$

If $\tau < \pi$, we infer from the case $\tau = \pi$ of what we have just proved that

$$\sum_n |a_n|^2 \geqq \frac{\pi}{10} \cdot \frac{\gamma}{2\pi} \int_{-\frac{\pi}{\gamma}}^{\frac{\pi}{\gamma}} |f(t)|^2 \, dt \geqq \frac{\tau}{10} \cdot \frac{1}{2T} \int_{-T}^{T} |f(t)|^2 \, dt.$$

The two inequalities together establish the theorem with $\mathfrak{a}(\tau) = \frac{1}{10} \min(\pi, \tau)$.

5. We pass now to the application of Theorem 1 to the analogue of the 'high indices' Tauberian theorem of Hardy and Littlewood.

**Theorem 3.** *Suppose that the Dirichlet's series* (1) *and the function $f(s)$ defined by it satisfy the following conditions:*

(i) $\lambda_n - \lambda_{n-1} \geqq \gamma > 0 \qquad (n > n_0)$;

(ii) *the series is convergent for $\sigma > 0$;*

(iii) *for some fixed* $T > \pi/\gamma$, $f(s)$ *is bounded in the rectangle* $0 < \sigma \leqq 1$, $-T \leqq t \leqq T$, *or, more generally, the mean value*

$$\frac{1}{2T}\int_{-T}^{T} |f(\sigma + ti)|^2\, dt \tag{12}$$

*is bounded as* $\sigma \to +0$;

(iv) $f(s)$ *is regular at* $s = 0$, *or, more generally, the function*

$$\frac{f(s) - f(0)}{s} \tag{13}$$

*(suitably defined on the line* $\sigma = 0$*) is continuous in the region* $\sigma \geqq 0$, $-\delta \leqq t \leqq \delta$, *for some* $\delta > 0$.

*Then* $\sum_1^\infty a_n$ *converges to the sum* $f(0)$.

In proving the theorem we may suppose (by considering $f(s) - \sum_1^{n_0-1} a_n e^{-\lambda_n s}$ in place of $f(s)$) that (i) holds for all $n > 1$. Writing $\gamma T = \pi + \varepsilon$ $(\varepsilon > 0)$ and applying Theorem 1 (5) to $f(\sigma + ti)$ with a fixed $\sigma > 0$, we obtain (since the Dirichlet's series is uniformly convergent over the range $-T \leqq t \leqq T$ for this $\sigma$)

$$\sum_1^\infty |a_n|^2 e^{-2\lambda_n \sigma} \leqq \frac{\mathfrak{A}(\varepsilon)}{2T}\int_{-T}^{T} |f(\sigma + ti)|^2\, dt.$$

Making $\sigma \to +0$, we deduce, by (iii), that $\sum |a_n|^2$ is convergent. Thus (i), (ii), (iii) imply that $a_n \to 0$ as $n \to \infty$. This is equivalent, by (i), to

$$a_n = o\left(\frac{l_n - l_{n-1}}{l_n}\right) \qquad (l_n = e^{\lambda_n}).$$

Introducing now the condition (iv), we can complete the proof by an appeal to either Theorem 1 $(l)$ or Theorem 3 $(l)$ (i) of my paper **1**. For the conditions of these theorems are satisfied with $\theta = 1$, $\Theta = 0$ [in 1 $(l)$], an arbitrarily small $K$ [in 3 $(l)$], and $\delta$ in place of $T$.

5. 1. If we compare Theorem 3 with earlier theorems of the same general character (e. g. Theorems 1 $(l)$ and 3 $(l)$ of the paper just quoted), we see that its distinguishing features are the presence of the condition (i) (which is equivalent to $l_n/l_{n-1} \geqq e^\gamma > 1$) and the absence of any explicit restriction on $a_n$. (The condition on $f(s)$ has been separated into the general condition (iii) and the local condition (iv) merely for convenience and greater generality.) The earlier theorems embodied non-trivial conclusions, of '$O$' type (in general), for every $T > 0$, which in some cases [e. g. in 3 $(l)$] merged into '$o$' conclusions when $T \to \infty$. Here, however,

there is an abrupt change as $T$ passes through the critical value $\pi/\gamma$. The theorem breaks down completely if $T$ falls below this value (as is obvious from the example of § 3, in which $|a_n| \sim n^{k-1}/(k-1)!$ as $n \to \infty$), but a '$o$' result appears as soon as $T$ exceeds it.

5. 2. With a view to possible generalisations of Theorem 3 we note that the proof may be based on the inequality (4) of Theorem 1 in place of the inequality (5). We leave the condition (iv) out of account in the first instance, and concentrate our attention on the proposition (established in the course of the above proof) that the conditions (i), (ii), (iii) imply that $a_n \to 0$; and in order to present the argument in its simplest form we shall first suppose (iii) replaced by the more special hypothesis that $f(s)$ (suitably defined on the line $\sigma = 0$) is continuous (and therefore uniformly continuous) in the rectangle $0 \leqq \sigma \leqq 1$, $-T \leqq t \leqq T$ (for some fixed $T > \pi/\gamma$). In this case

$$(14) \qquad \frac{1}{2T} \int_{-T}^{T} |f(\sigma + ti) - f(\sigma' + ti)|\, dt \to 0 \text{ as } \sigma, \sigma' \to +0,$$

rom which we deduce, applying (4) to the function

$$f(\sigma + ti) - f(\sigma' + ti) = \sum_{1}^{\infty} a_n (e^{-\lambda_n \sigma} - e^{-\lambda_n \sigma'}) e^{-\lambda_n ti},$$

and putting $\sigma = \lambda_n^{-1}$, $\sigma' = 2\lambda_n^{-1}$, that $|a_n(e^{-1} - e^{-2})| \to 0$ as $n \to \infty$, i. e. $a_n \to 0$ as required. The argument may be extended to the case in which $f(s)$ is bounded in $0 < \sigma \leqq 1$, $-T \leqq t \leqq T$, by an appeal to the well known theorem of Fatou on the existence of boundary values of bounded analytic functions. The analogy of power series suggests that the conclusion would remain valid if (iii) were replaced by the boundedness of

$$\frac{1}{2T} \int_{-T}^{T} |f(\sigma + ti)|\, dt$$

(for a fixed $T > \pi/\gamma$) as $\sigma \to +0$ (a wider condition in virtue of Schwarz's inequality). This would certainly be so if from this hypothesis we could infer (14), or even a similar relation with $T$ replaced by a $T'$ between $\pi/\gamma$ and $T$. Such an inference seems highly plausible, but I am not able to quote any known result which is immediately applicable.

Generalisations of the condition (iv), of a 'real variable' character, have been considered in similar contexts by other writers[3]), and will not be further discussed here.

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[3]) M. Riesz, 1 (Satz II); Karamata, 2.

6. Instead of generalising the conditions of Theorem 3 we might specialise them for the sake of simplicity of statement, for example by replacing (iii) and (iv) by the continuity of the function (13) for $\sigma \geqq 0$, $-T \leqq t \leqq T\,(T > \pi/\gamma)$, or by the still more special hypothesis that $f(s)$ is regular along the stretch $-\pi/\gamma \leqq t \leqq \pi/\gamma$ of the line $\sigma = 0$. The latter might seem to be the natural 'practical' condition, but if it is adopted the conclusion of the theorem, namely that the Dirichlet's series is convergent at the point $s = 0$, can be replaced by the stronger assertion that this point actually lies in the half-plane of convergence. This is an immediate consequence of the following known result — a particular case of an important theorem due to Pólya[4]):

*If*

$$\varliminf_{n\to\infty} (\lambda_n - \lambda_{n-1}) \geqq \gamma > 0,$$

*every closed interval of length* $2\pi/\gamma$ *on the line of convergence of the Dirichlet's series* (1) *contains a singular point of the function* $f(s)$ *defined by the series.*

It is of interest to note that Pólya's theorem (in the special form quoted) may be deduced very simply from Theorem 1[5]). Assuming the result false, we may suppose (by a linear change of the variable $s$) that $\sigma = 0$ is the line of convergence of the series (1), and that $f(s)$ has no singularity on the segment $-\pi/\gamma \leqq t \leqq \pi/\gamma$ of this line. Next we may suppose that $\varliminf (\lambda_n - \lambda_{n-1}) > \gamma$ (otherwise take a smaller $\gamma$ near enough to the old to ensure that $f(s)$ remains regular along the slightly extended segment $-\pi/\gamma \leqq t \leqq \pi/\gamma$ of $\sigma = 0$). Then we shall have $\lambda_n - \lambda_{n-1} \geqq \gamma$ for all sufficiently large $n$, and we may suppose (by subtracting a finite number of terms) that this holds for all $n > 1$. And we may now take $\gamma = 1$ (by a change of variable).

Having made these simplifying assumptions, we choose $\varepsilon > 0$ so small that $f(s)$ is regular in the closed rectangle $-3\varepsilon \leqq \sigma \leqq 1 + 3\varepsilon$, $|t| \leqq \pi + 4\varepsilon$. Let $M$ be the upper bound of $|f(s)|$ in this rectangle. Consider $f^{(q)}(s)$, where $q$ is any positive integer or zero. For $\sigma > 0$ we have the convergent expansion

$$f^{(q)}(s) = \sum_1^\infty (-1)^q\, a_n\, \lambda_n^q\, e^{-\lambda_n s}.$$

For $0 < \sigma \leqq 1$, $|t| \leqq \pi + \varepsilon$, we have, $C$ denoting the circle $|z| = 3\varepsilon$,

$$|f^{(q)}(s)| = \left| \frac{q!}{2\pi i} \int_C \frac{f(s+z)}{z^{q+1}}\, dz \right| \leqq \frac{q!\, M}{(3\varepsilon)^q}.$$

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[4]) Pólya, 1; V. Bernstein, 1, 140.

[5]) The proof which follows has some features in common with the proof given by Carlson and Landau (1) of Fabry's gap theorem and its generalisations.

Applying Theorem 1 [(4) or (5)] to $f^{(q)}(\sigma + t\,i)$, we infer that

$$|(-1)^q a_n \lambda_n^q e^{-\lambda_n \sigma}| \leqq \mathfrak{A}(\varepsilon) \frac{q!\,M}{(3\varepsilon)^q} \qquad (0 < \sigma \leqq 1),$$

whence, making $\sigma \to 0$ and rearranging,

$$|a_n| \frac{(2\lambda_n \varepsilon)^q}{q!} \leqq \mathfrak{A}(\varepsilon)\, M \left(\frac{2}{3}\right)^q.$$

Summing from $q = 0$ to $\infty$, we deduce

$$|a_n|\, e^{2\lambda_n \varepsilon} \leqq 3\,\mathfrak{A}(\varepsilon)\, M.$$

Hence

$$a_n e^{\lambda_n \varepsilon} = O(e^{-\lambda_n \varepsilon}) = O(e^{-n\varepsilon})$$

as $n \to \infty$, and the series (1) is convergent for $s = -\varepsilon$, contrary to the hypothesis that $\sigma = 0$ is the line of convergence.

7. The inequalities of Theorems 1 and 2 extend at once (in virtue of the 'approximation theorems'[6]) to the class of uniformly almost periodic functions with Fourier exponents $\Lambda_n$ satisfying

$$|\Lambda_m - \Lambda_n| \geqq \gamma > 0 \qquad (m \neq n). \tag{15}$$

This class includes the class of purely periodic continuous functions of period $2\pi/\gamma$, and Bochner has shown that many of the special properties of periodic functions and their Fourier series extend to the larger class[7]). To these we may add the following simple result, in which the number $2\pi/\gamma$ plays a part corresponding to a period of a periodic function.

**Theorem 4.** *A uniformly almost periodic function $f(t)$ which is not identically 0, and whose Fourier exponents satisfy* (15), *cannot vanish throughout an interval of length $2\pi/\gamma$.*

Assuming the theorem false, we may take $\gamma = 1$ and suppose that $f(t) = 0$ for $-\pi < t < \pi$. Then, denoting by $A_n$ the Fourier coefficients of $f(t)$, and taking $\varepsilon > 0$, we have, by Theorem 1 (5) (extended, as explained, to almost periodic functions),

$$\sum_n |A_n|^2 \leqq \frac{\mathfrak{A}(\varepsilon)}{2(\pi+\varepsilon)} \int_{-\pi-\varepsilon}^{\pi+\varepsilon} |f(t)|^2\,dt = \frac{\varepsilon\,\mathfrak{A}(\varepsilon)}{2(\pi+\varepsilon)} \cdot \frac{1}{\varepsilon} \int_{\pi}^{\pi+\varepsilon} (|f(t)|^2 + |f(-t)|^2)\,dt.$$

Making $\varepsilon \to 0$ and supposing $\mathfrak{A}(\varepsilon)$ chosen so that $\varepsilon\,\mathfrak{A}(\varepsilon)$ tends to an absolute constant $\mathfrak{A}$ (as we may by Theorem 1), we deduce, since $f(t)$ is continuous,

$$\sum_n |A_n|^2 \leqq \frac{\mathfrak{A}}{2\pi}(|f(\pi)|^2 + |f(-\pi)|^2) = 0,$$

which is impossible since $f(t)$ is not identically 0.

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[6]) Besicovitch, 1, 29—31 and 20—21 (11⁰).

[7]) Bochner, 1; Besicovitch, 1, 38—42.

We can avoid the introduction of $\varepsilon$ by adapting the last proof of (5) in § 2 instead of appealing to the result itself. The formal relations of this proof remain valid when $\varepsilon = 0$. Thus $K_0(0) = 0$, and we have, by (11),

$$0 = \int_{-\infty}^{\infty} k_0(t)\,|f(t)|^2\,dt, \tag{16}$$

where

$$k_0(t) = \frac{1+\cos t}{\pi^2 - t^2}$$

(the extension from trigonometrical polynomials to $f(t)$ by 'approximation' presenting no difficulty since $|k_0(t)|$ is integrable over $(-\infty, \infty)$). Now $k_0(t) \leqq 0$ for $|t| > \pi$, with equality only at the points $t = \pm 3\pi$, $\pm 5\pi, \ldots$. If $f(t)$ vanished for $-\pi < t < \pi$ but not identically, the integral on the right of (16) would therefore be negative and not zero.

8. We turn now to the question (raised in § 3) of the possibility of extending Theorems 1 and 3 to the critical case $T = \pi/\gamma$, with $\mathfrak{A}$ in place of $\mathfrak{A}(\varepsilon)$ in (4) and (5). It will be shown by an example that no such extension of (5) is possible. But I have not found the answer to the question for the inequality (4), and this implies (after what was said in 5.2) a similar uncertainty with regard to Theorem 3.

8.1. Before giving an actual example in connection with (5) we shall examine systematically how the methods of § 2 break down when we try to adapt them to the critical case. If $k(t)$ and $K(u)$ are connected by (6) (which implies that $k(t+T)$ and $e^{uTi}K(u)$ are similarly related), and if $\gamma T = \pi$, Poisson's formula[8]) gives (at any rate formally)

$$2T\sum_{-\infty}^{\infty} k(2nT+T) = \sum_{-\infty}^{\infty} (-1)^n K(n\gamma). \tag{17}$$

Now the first method of § 2 requires $k(t) = 0$ for $|t| \geqq T$, and

$$\sum_{n \neq 0} |K(\mu_n)| < \theta\,|K(0)| < |K(0)| \tag{18}$$

for all sets of real numbers $\ldots \mu_{-1}, \mu_0, \mu_1, \mu_2, \ldots$ satisfying $\mu_0 = 0$, $\mu_n - \mu_{n-1} \geqq \gamma$, where $\theta$ is independent of the set $(\mu_n)$. But the first of these conditions implies that the left hand side of (17) is 0 and therefore that

$$K(0) = \sum_{n \neq 0} (-1)^{n-1} K(n\gamma), \tag{19}$$

which in turn implies

$$|K(0)| \leqq \sum_{n \neq 0} |K(n\gamma)| \tag{20}$$

and therefore contradicts (18) for the particular set $\mu_n = n\gamma$. The second method of § 2 (i. e. the last proof of (5)) requires $k(t) \leqq 0$ for

[8]) See, e. g., Bochner, 2, 33 (3).

$|t| \geqq T$, $K(u) = 0$ for $|u| \geqq \gamma$, and $K(0) > 0$. The first two of these conditions imply, by (17), that

$$K(0) = 2T \sum_{-\infty}^{\infty} k(2nT + T) \leqq 0, \tag{21}$$

which contradicts the third condition, so that this method also fails. We have tacitly assumed $k(t)$ and $K(u)$ to be continuous, and it might be supposed that the scope of the methods could be extended by the introduction of discontinuities. But it is easy to show that our requirements preclude such a possibility and ensure the essential validity of the formal reasoning.

The preceding observations provide the clue to the choice of the functions $k(t)$ and $K(u)$ in § 2. We come nearest to satisfying our requirements in the critical case $\gamma T = \pi$ when the sign of equality holds in (20) (for the first method), and in (21) (for the second). For this it is necessary and sufficient that all terms in (19) should be of the same sign (in the first case), and that $k(t)$ should vanish for $t = \pm T$, $\pm 3T$, $\pm 5T, \ldots$ (in the second). To obtain suitable functions for use in § 2 we first choose functions $k(t)$ and $K(u)$ satisfying the above conditions for the critical case, and then modify them slightly by the introduction of $\varepsilon$; we must, however, be careful to arrange that (18) can be proved for *all* admissible sets $(\mu_n)$.

8.2. Coming now to the construction of an actual example to show that (5) cannot be improved in the sense suggested, let

$$g(z) = \sum_{0}^{\infty} b_m z^m$$

be any function regular for $|z| < 1$, and take

$$f(t) = e^{\frac{\lambda}{2} t i} g(r e^{t i}) \pm e^{-\frac{\lambda}{2} t i} g(r e^{-t i}),$$

where $0 < r < 1$ and $\lambda > 0$. Then

$$f(t) = \sum_{0}^{\infty} b_m r^m e^{\left(m + \frac{\lambda}{2}\right) t i} \pm \sum_{0}^{\infty} b_m r^m e^{-\left(m + \frac{\lambda}{2}\right) t i},$$

where the series converge uniformly for all real $t$ ($r, \lambda$ fixed). If $\lambda \geqq 1$, $f(t)$ is of the form $\sum_{-\infty}^{\infty} a_n e^{-\lambda_n t i}$ with $\lambda_n - \lambda_{n-1} \geqq 1$, and the proposed refinement of (5) would therefore imply (for $0 < r < 1$)

$$\frac{1}{\pi} \int_{-\pi}^{\pi} |g(r e^{t i})|^2 \, dt \leqq \frac{\mathfrak{A}}{2\pi} \int_{-\pi}^{\pi} \left| e^{\frac{\lambda}{2} t i} g(r e^{t i}) \pm e^{-\frac{\lambda}{2} t i} g(r e^{-t i}) \right|^2 dt, \tag{22}$$

the left hand side being equal to $2 \Sigma |b_m|^2 r^{2m}$ $(= \Sigma |a_n|^2)$.

Take $g(z) = (1+z)^{-\alpha}$, where $\alpha > 0$ and the principal value is to be taken. Then, for $-\pi < t < \pi$,

$$g(r e^{\pm t i}) \to (2\cos\tfrac{1}{2}t)^{-\alpha} e^{\mp \frac{\alpha}{2} t i} \quad \text{as } r \to 1,$$

$$|g(r e^{\pm t i})|^2 = \{(1-r)^2 + 4r\cos^2\tfrac{1}{2}t\}^{-\alpha} < (\cos\tfrac{1}{2}t)^{-2\alpha} \quad (\tfrac{1}{4} < r < 1).$$

If $2\alpha < 1$, this majorant is integrable in $(-\pi, \pi)$ and we may make $r \to 1$ under the integral signs in (22). This gives

$$\int_{-\pi}^{\pi} (\cos\tfrac{1}{2}t)^{-2\alpha}\, dt \leqq 2\,\mathfrak{A} \int_{-\pi}^{\pi} (\cos\tfrac{1}{2}t)^{-2\alpha} \begin{pmatrix}\cos \\ \sin\end{pmatrix}^2 \frac{\lambda-\alpha}{2} t\, dt \quad (0 < \alpha < \tfrac{1}{2},\ \lambda \geqq 1).$$

In this take the upper alternative with $\lambda = \alpha + 1$ $(> 1)$; we get

$$\int_{-\pi}^{\pi} (\cos\tfrac{1}{2}t)^{-2\alpha}\, dt \leqq 2\,\mathfrak{A} \int_{-\pi}^{\pi} (\cos\tfrac{1}{2}t)^{2-2\alpha}\, dt \qquad (0 < \alpha < \tfrac{1}{2}).$$

But this is impossible, since the integral on the right is bounded, while that on the left is unbounded, when $\alpha \to \frac{1}{2}$. (We could obtain a similar contradiction by taking the lower alternative with $\lambda = \alpha + 2$.) A slight variation of the procedure would be to take $\alpha = \frac{1}{2}$, $\lambda = \frac{3}{2}$ (or $\frac{5}{2}$) and show that the integral on the right of (22) is bounded as $r \to 1$, while that on the left is asymptotically $-2\log(1-r)$.

We may note (returning to the general $g$) that (22) is equivalent to

$$(23) \qquad \sum_{m=0}^{\infty} |\beta_m|^2 \leqq \mathfrak{A}\Big( \sum_{m=0}^{\infty} |\beta_m|^2 \pm \frac{\sin\lambda\pi}{\pi} \sum_{m,\,n=0}^{\infty} \frac{\Re(\beta_m \bar{\beta}_n)}{m+n+\lambda} \Big),$$

where $\beta_m = (-1)^m b_m r^m$. The fact that we can falsify (22) by taking $\pm \sin\lambda\pi$ near to $-1$ and choosing $g(z)$ and $r$ appropriately is thus seen to be connected with the fact that $\pi$ is the best possible constant in Hilbert's inequality; and the above examples are variants of the standard example used to establish the latter fact[9]).

8.3. It may be mentioned, in conclusion, that, although the truth of Theorem 3 in the critical case $T = \pi/\gamma$ remains in doubt, there is at any rate no such complete breakdown as when $T < \pi/\gamma$, for the conditions still impose definite limitations on the order of magnitude of $a_n$. I state without proof some rather fragmentary results in this direction.

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[9]) Hardy, Littlewood, and Pólya, **1**, 232—234. Since, in the example in question, the sums of the series are large compared with the individual terms, the value of $\lambda$ in $m+n+\lambda$ is immaterial so long as it lies between fixed positive bounds.

*If the conditions* (i), (ii), (iii) *of Theorem* 3 *are satisfied, but with* $T = \pi/\gamma$ *in* (iii), *then*

$$\sum_{\lambda_n \leqq \omega} |a_n|^2 = O(\omega) \quad as \quad \omega \to \infty; \tag{24}$$

$$a_n = O\{(\log \lambda_n)^{1/2}\} \quad as \quad n \to \infty; \tag{25}$$

$$\sum_{2}^{\infty} |a_{n-1} + a_n|^2 < \infty; \tag{26}$$

$$\sum_{\lambda_n - \lambda_{n-1} \geqq (1+\delta)\gamma} (|a_{n-1}|^2 + |a_n|^2) < \infty \qquad (\delta > 0). \tag{27}$$

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